

YU YUAN

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RESEARCH INTERESTS

Physically-consistent Generation, Physics-informed Vision, Physical AI, World Models, Image/Video Synthesis, Generative Modeling, Computational Photography, Diffusion Models, Image/Video Restoration.

EDUCATION

Purdue University

PhD Student in Electrical and Computer Engineering

Advisor: Stanley H. Chan

West Lafayette, IN, US

08/2023 – 06/2027 (expected)

Shanghai Jiao Tong University

MS in Aeronautical and Astronautical Science and Technology

Shanghai, China

09/2020 – 06/2023

Shanghai Jiao Tong University

BEng in Aeronautics and Astronautics Engineering

Minor Degree in Administration Management

Shanghai, China

09/2016 – 06/2020

10/2018 – 06/2020

SKILLS

Languages: English (Fluent), Chinese (Native)

Programming: Python, PyTorch, MATLAB, HTML, JavaScript, C/C++ (Basic)

Technologies: Git, LaTeX, Linux, Slurm, Markdown, NVIDIA Jetson, Adobe Photoshop, Adobe Premiere, Adobe Audition

Others: Photography, Fishing, Drones, Rowing

SELECTED PUBLICATIONS

First Author. NewtonGen: Physics-Consistent and Controllable Text-to-Video Generation via Neural Newtonian Dynamics. arXiv 2025 [Under Review]

- Built a physics-consistent and controllable text-to-video framework that explicitly incorporates learnable dynamics into the generation process.
- Introduced Neural Newtonian Dynamics, which models different dynamics via unified neural ODEs.

First Author. Generative Photography: Scene-Consistent Camera Control for Realistic Text-to-Image Synthesis. CVPR 2025 [Highlight & Demo]

- Introduced a new text-to-image generation paradigm for photography with an understanding of camera physics.
- Produced significantly more scene-consistent photorealistic images than state-of-the-art text-to-image models such as Stable Diffusion 3 and FLUX.

Co-Author. Learning Phase Distortion with Selective State Space Models for Video Turbulence Mitigation. CVPR 2025 [Highlight]

- Performed video turbulence restoration by integrating the Mamba model with generative models, surpassing state-of-the-art methods in quality and inference speed.

First Author. iHDR: Iterative HDR Imaging with Arbitrary Number of Exposures. ICIP 2025

- Proposed a novel high dynamic range (HDR) imaging framework capable of handling a flexible number of inputs.
- Utilized side information and attention mechanisms to effectively suppress artifacts during the fusion process.

Co-Author. Astrophotography Turbulence Mitigation via Generative Models. ICIP 2025

- Established the first dataset dedicated to astronomical image turbulence restoration, comprising over 135K images.
- Presented a novel architecture integrating a generative prior for structural preservation with a restoration model.

Co-Author. RCMixer: Radar-Camera Fusion Based on Vision Transformer for Robust Object Detection. JVCJ 2024

- Proposed an end-to-end target detection network architecture based on radar-camera feature-level fusion.

First Author. Learning to Kindle the Starlight. arXiv 2022

- Constructed the first star field image enhancement benchmark consisting of 355 real-shot and 854 semi-synthetic image pairs.
- Proposed the first star field image enhancement approach based on conditional denoising diffusion probabilistic models (DDPM), which outperformed state-of-the-art low-light image enhancement algorithms on different tasks.

First Author. Multimodal Image Fusion based on Hybrid CNN-Transformer and Non-local Cross-modal Attention. arXiv 2022

- Designed a non-local cross-modal attention mechanism to capture both local and global dependencies by calculating associations between features.

LEADING PROJECTS

Physics-driven Video Understanding and Generation. 2025

Time Disentanglement for Latent Space Motion Trajectory Generation. 2025

Generative Astronomical Image Restoration Through Atmospheric Turbulence. 2024

Video Restoration Classifier. 2024

A Underwater Binocular Multi-exposure Imaging System. 2022

Ultra-wideband-based Cooperative Multi-UAV Positioning Technology. 2021

A Modular Custom Aviation Model Design for Tyros. 2020

PATENT

A trajectory solving and alignment method based on inertial measurement unit and ultra-short baseline positioning sensors for autonomous underwater robots. China Patent. CN 114993313 B.

AWARDS & COMPETITIONS

Ivy League Scholarship for Exceptional Students (Innovation)	Top 1%
Outstanding Graduate of Shanghai Jiao Tong University	Top 15%
Shanghai International Creators Competition: Special Competition of Drones	First Prize
DELL AI for Social Innovation Competition: AI/ADAS of Intelligent Car	National Second Prize

OTHER ACTIVITIES

Research Assistant , Purdue University	2023 – 2025
Teaching Assistant , Automatic Control Theory, Shanghai Jiao Tong University	2021
Workshop Co-organizer , WACV GAIP	
Reviewer , CVPR, CVPRW, ICCV, WACV, ICIP, ICASSP	
IEEE Graduate Student Member	